

Lingchen Meng

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RESEARCH INTERESTS

Systems Security: Cyber-Physical Systems Security; Computer System Security

AI/Robotics Security: Embodied AI Security

EDUCATION

Ph.D. in Computer Science

Aug 2025 – Present

Advisor: Dr. Ning Zhang

Washington University in St. Louis, St. Louis, MO

B.S. in Computer Science

Aug 2022 – May 2025

Washington University in St. Louis, St. Louis, MO

GPA: 3.98/4.00

Honors: Summa Cum Laude

HONORS

- Dean's Select PhD Fellowship
 - Tau Beta Pi Engineering Honor Society Member
 - Dean's List, Fall 2022 – Spring 2025
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Ongoing Research

Physical Adversarial Examples in ML Perception Systems

Researcher

Fall 2025 - Present

- Modeled physical adversarial examples as an end-to-end physical-to-digital attack pipeline
- Surveyed 130+ prior works and analyzed limitations in existing robustness evaluations
- Built a controlled experimental platform to evaluate how physical and sensing factors affect adversarial attack reliability

Measuring Impact Hardness of AI Command Errors in Embodied Control Stacks

Researcher

Spring 2026 - Present

- Perform static analysis on AI-enabled control stack, such as self-driving and humanoid.
 - Investigate the VLA model and the control logic of the Unitree G1 Humanoid
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Work Experience

AR Admission Letter of The High School Affiliated to Renmin University of China

Leader, Developer

Beijing, China

Summer 2023, Winter 2023

- Held weekly meetings to check in with teammates, assign work and requirements
- Established AR environment and designed interactive elements
- Managed the application distribution for the iOS system and kept track of user data

Introduction to Computer Engineering & Introduction to Computer Security

Teaching Assistant

Washington University

Fall 2023 - Fall 2024

- Held in-person office hours and answered students' questions
 - Helped students with their in-class studio
 - Graded assignments and exams
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Technical Project

Command & Control Malware Development

Project Developer

Washington University

Fall 2023

- Developed a program to connect to another computer and operate many commands
- Implemented functionalities to control mouse movement, listen to keyboard input, capture microphone input, and monitor the victim's screen
- Decrypted Chrome and Edge saved passwords and send them back to the server
- Able to check whether it is in debug mode or being analyzed to prevent reverse engineering

Reverse Engineering Camera Slider APK

Project Developer

Washington University

Summer 2024

- Used both static and dynamic analyses to gather information on APK file
- Based on the behavior, reversed the BLE protocol and designed script to precisely control slider

Reverse Engineering Minesweeper Game

Project Developer

Washington University

Spring 2024

- Used both static and dynamic analyses to gather information on exe file
- Analyzed assembly code from the executable
- Modified assembly code to customize finish time, stop the clock at a particular time, and show the position of bombs immediately after the game starts

Memory Allocation Simulation

Project Developer

Washington University

Fall 2023

- Implemented a heap simulator supporting heap initialization, allocation, free, reallocation, contiguous allocation, extend heap, and coalesce.
- Used explicit free list to increase throughput and N-th fit to increase space utilization.

Simple Calendar Website Development

Project Developer

Washington University

Fall 2023

- Developed a user registration and login system and stored the usernames and hashed and salted passwords in SQL database
- Implemented functionalities for users to add, delete, and edit events, which are displayed on the calendar with different colors based on the event category
- Website reminds users 5 minutes before the starting time of an event

Simple Chat Room Website Development

Project Developer

Washington University

Fall 2023

- Developed functionalities for users to create public or private rooms with password protection
- Implemented chat functionalities for public and private messages within the room
- Administrator can assign or remove co-admin, mute or unmute, kick, and ban users.

Communication with AES-GCM Encryption Project

Project Developer

Washington University

Fall 2023

- Developed a messaging system based on socket between client and server with AES-GCM encryption
- Ensured both confidentiality and integrity with this encryption

File Managing System Development

Project Developer

Washington University

Spring 2023

- Developed a file management system similar to a Linux
- Supported commands: cat, copy, display, ls, remove, rename, and touch
- Implemented password protection for files

Skills

Programming Skills:

C/C++, Python, Java, JavaScript, HTML, CSS, PHP, MySQL, R, MATLAB, Assembly, LaTeX, Linux Kernel

Software:

Visual Studio, VS Code, AWS EC2, MATLAB, gdb, Wireshark, IDA Pro, Immunity Debugger, PE-View, Metasploit

Languages:

English, Chinese